Phishing Awareness Training

Learning outcomes

In this module you will learn the following:

- What Phishing is.
- Recognising phishing emails and fake websites.
- Tactics used by attackers.
- How one can prevent these attacks.

Introduction

Phishing is one of the most common cyberattacks used by criminals to steal sensitive information which includes usernames, passwords, banking details, and personal data. Attackers use deceptive emails, messages, and websites that appear legitimate to trick users into revealing confidential information[1]. As more people rely on the internet for communication, shopping, and banking, understanding phishing attacks and knowing how to identify them has become essential for staying safe online. This module aims to raise awareness about phishing, explain how attackers operate, and provide practical methods for identifying and preventing phishing attacks.

What is Phishing

This is a social engineering method that attackers make use to trick users into doing what they want leading to the user leaking sensitive and personal information which leads to identity theft. This method is highly used due to an increase in number of people on the internet who are sharing some of their personal information online, this then gives attackers some information to start with their attacks on the user[2][3].

How recognising phishing emails and fake websites.

- Fake emails and websites often contain warning signs that users can identify. Phishing emails may create a sense of urgency by claiming that an account will be suspended or that immediate action is required. They may also contain spelling and grammatical mistakes, suspicious attachments, or requests for personal information[4].
- When examining a website, users should carefully check the URL to ensure it matches the legitimate website. Fraudulent websites often use misspelled domain names or additional characters that make them appear genuine at first glance. Users should also look for HTTPS in the address bar and avoid websites that display security certificate warnings. If a website appears suspicious, users should verify its authenticity before entering any personal information[2][5].

Tactics used by attackers

- There are a lot of tactics that attackers utilises such as presenting a high credible website presence to the victims[1]
- Attackers send fake email which have links to the user, and when the user clicks on the link it will take them to a site that is a replica of legitimate companies that the user are affiliated with[2].
- They exploit the lack of digital or cyber knowledge from the users, due to the user not being able to comprehend the syntax used for domain naves thus they know users are likely to fall victim if they are provided with a high credible website because they can not distinguish between legit and fraudulent URL,s and forged email headers[6].

How one can prevent these attacks

- Users must pay attention to pop up warnings about fraudulent certificates that show up on a website and make sure to leave that website before providing information[1].
- Users must not confirm their personal information over the internet if they are not familiar with the sender[4].
- Users must limit the amount of personal information they share on the internet[4].
- Users must confirm their information via calls and secure websites they trust[6].
- Users must use different passwords for each account[4].

Activities

- 1. What is phishing?
  A. A type of antivirus software
  B. A social engineering attack used to steal information
  C. A computer programming language
  D. A network cable

- 2. Which of the following is a sign of a phishing email?
  A. Email from a trusted sender you know personally
  B. Correct spelling and grammar throughout
  C. Urgent requests for personal information
  D. A message you were expecting

- 3. What should you do if a website displays a certificate warning?
  A. Ignore the warning and continue
  B. Provide your information quickly
  C. Leave the website and verify its legitimacy
  D. Refresh the page repeatedly

- 4. Why do attackers create fake websites?
  A. To improve internet speed
  B. To collect users' sensitive information
  C. To update software
  D. To provide free services

- 5. Which practice helps protect you from phishing attacks?
  A. Using the same password for all accounts
  B. Sharing personal information online freely
  C. Verifying requests through trusted channels
  D. Clicking links from unknown senders

Test your understanding by examining sample emails and identify:

- signs of phishing.
- Suspicious links or attachments.
- Urgent or threatening language.
- Requests for personal information.

Answers to Q1-Q5: B,C,C,B,C

Summary

Phishing is a social engineering attack that aims to trick users into revealing sensitive information through fake emails, messages, and websites. Attackers use different tactics, including fake websites and fraudulent emails, to exploit users' trust and lack of cybersecurity awareness. By learning to identify suspicious emails, verifying website URLs, avoiding the sharing of personal information with unknown sources, and following safe online practices, users can significantly reduce their risk of becoming victims of phishing attacks. Cybersecurity awareness and vigilance are key to protecting personal and organizational information from phishing threats.

References

[1] M. Khonji, Y. Iraqi, and A. Jones, “Phishing detection: A literature survey,” *IEEE Communications Surveys and Tutorials*, vol. 15, no. 4, pp. 2091–2121, 2013, doi: 10.1109/SURV.2013.032213.00009.

[2] A. Aleroud and L. Zhou, “Phishing environments, techniques, and countermeasures: A survey,” *Comput. Secur.*, vol. 68, pp. 160–196, Jul. 2017, doi: 10.1016/J.COSE.2017.04.006.

[3] K. Jansson and R. Von Solms, “Phishing for phishing awareness,” *Behaviour & Information Technology*, vol. 32, no. 6, pp. 584–593, Jun. 2013, doi: 10.1080/0144929X.2011.632650.

[4] I. Fette, N. Sadeh, and A. Tomasic, “Learning to detect phishing emails,” *16th International World Wide Web Conference, WWW2007*, pp. 649–656, 2007, doi: 10.1145/1242572.1242660.

[5] R. Dhamija, J. D. Tygar, and M. Hearst, “Why phishing works,” *Conference on Human Factors in Computing Systems - Proceedings*, vol. 1, pp. 581–590, 2006, doi: 10.1145/1124772.1124861.

[6] Z. Alkhalil, C. Hewage, L. Nawaf, and I. Khan, “Phishing Attacks: A Recent Comprehensive Study and a New Anatomy,” *Front. Comput. Sci.*, vol. 3, p. 563060, Mar. 2021, doi: 10.3389/FCOMP.2021.563060/TEXT.